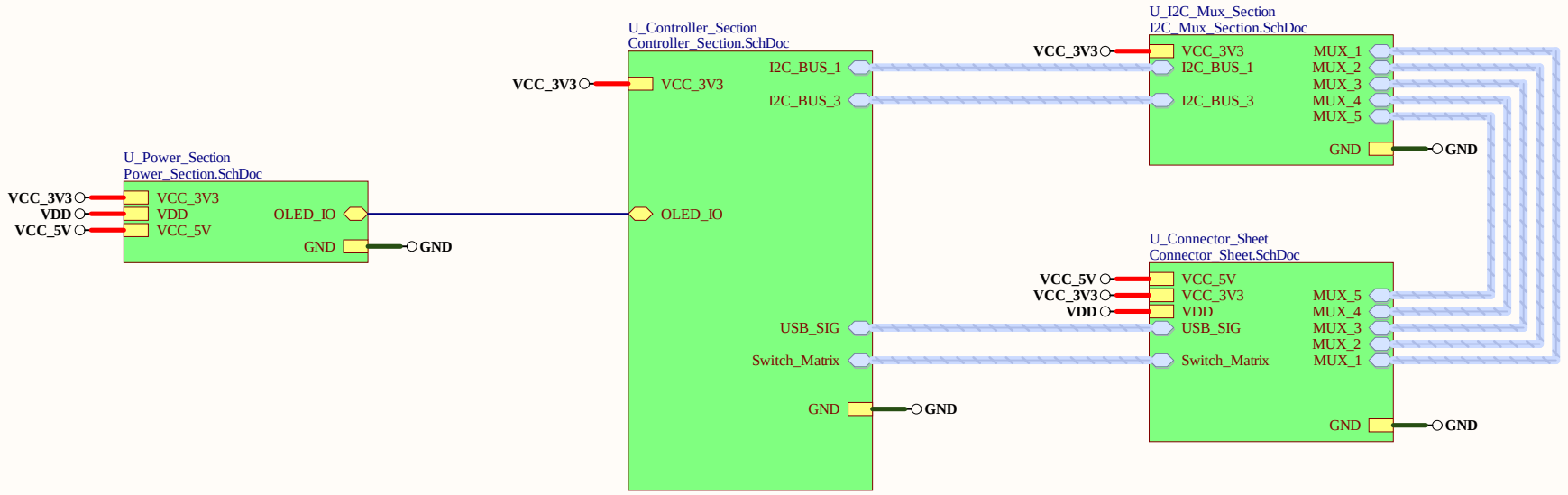
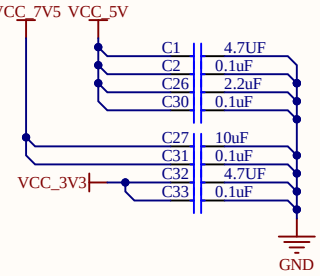
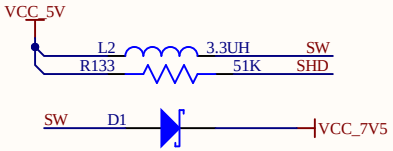

U_Power_Section
Power_Section.SchDoc
VCC_3V3
VDD
VCC_5V
VCC_3V3
VDD
OLED_IO
GND
GND
U_Controller_Section
Controller_Section.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_1
I2C_BUS_3
OLED_IO
USB_SIG
Switch_Matrix
GND
GND
U_I2C_Mux_Section
I2C_Mux_Section.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_1
I2C_BUS_3
MUX_1
MUX_2
MUX_3
MUX_4
MUX_5
GND
GND
U_Connector_Sheet
Connector_Sheet.SchDoc
VCC_5V
VCC_3V3
VDD
VCC_5V
VCC_3V3
VDD
USB_SIG
Switch_Matrix
MUX_5
MUX_4
MUX_3
MUX_2
MUX_1
GND
GND

OLED_IO OLED_IO

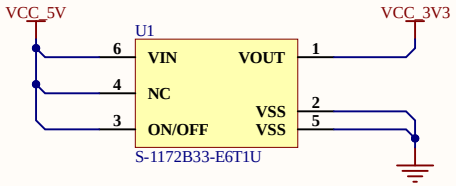
DECOUPLING CAPACITORS



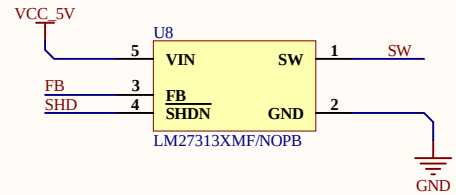
SUPPORTING CIRCUIT



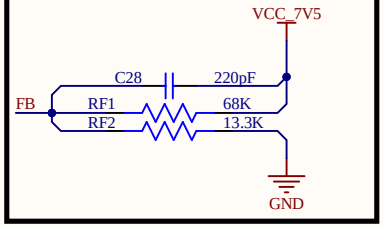
3.3V FIXED REGULATOR



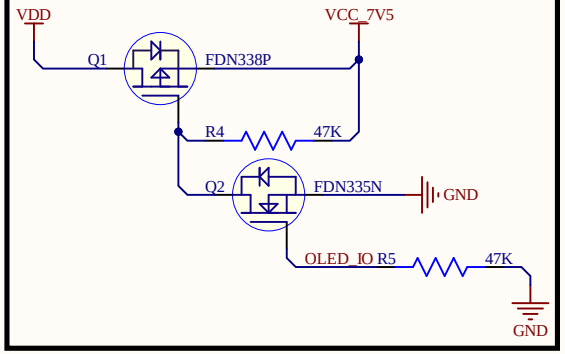
BOOST REGULATOR

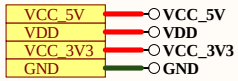


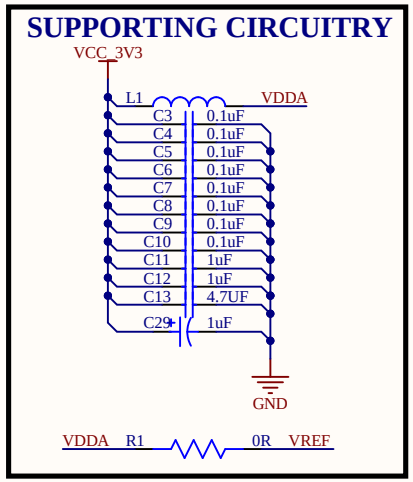
FEEDBACK NETWORK

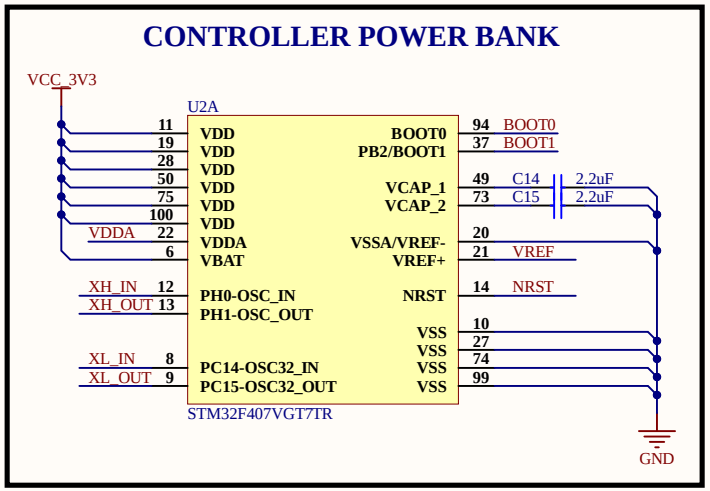


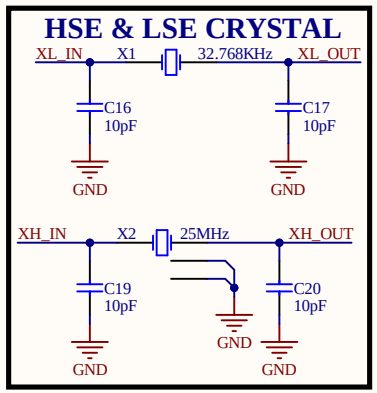
MOSFET CONTROL CIRCUIT

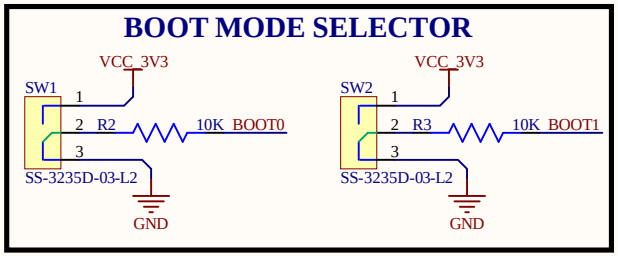


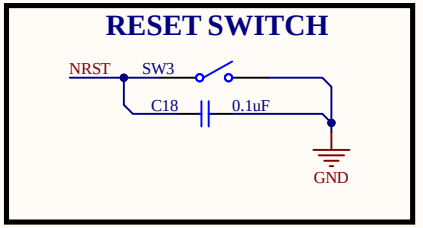



SUPPORTING CIRCUITRY
VCC_3V3
L1
VDDA
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 1uF
C12 1uF
C13 4.7UF
C29 1uF
GND
VDDA R1 0R VREF

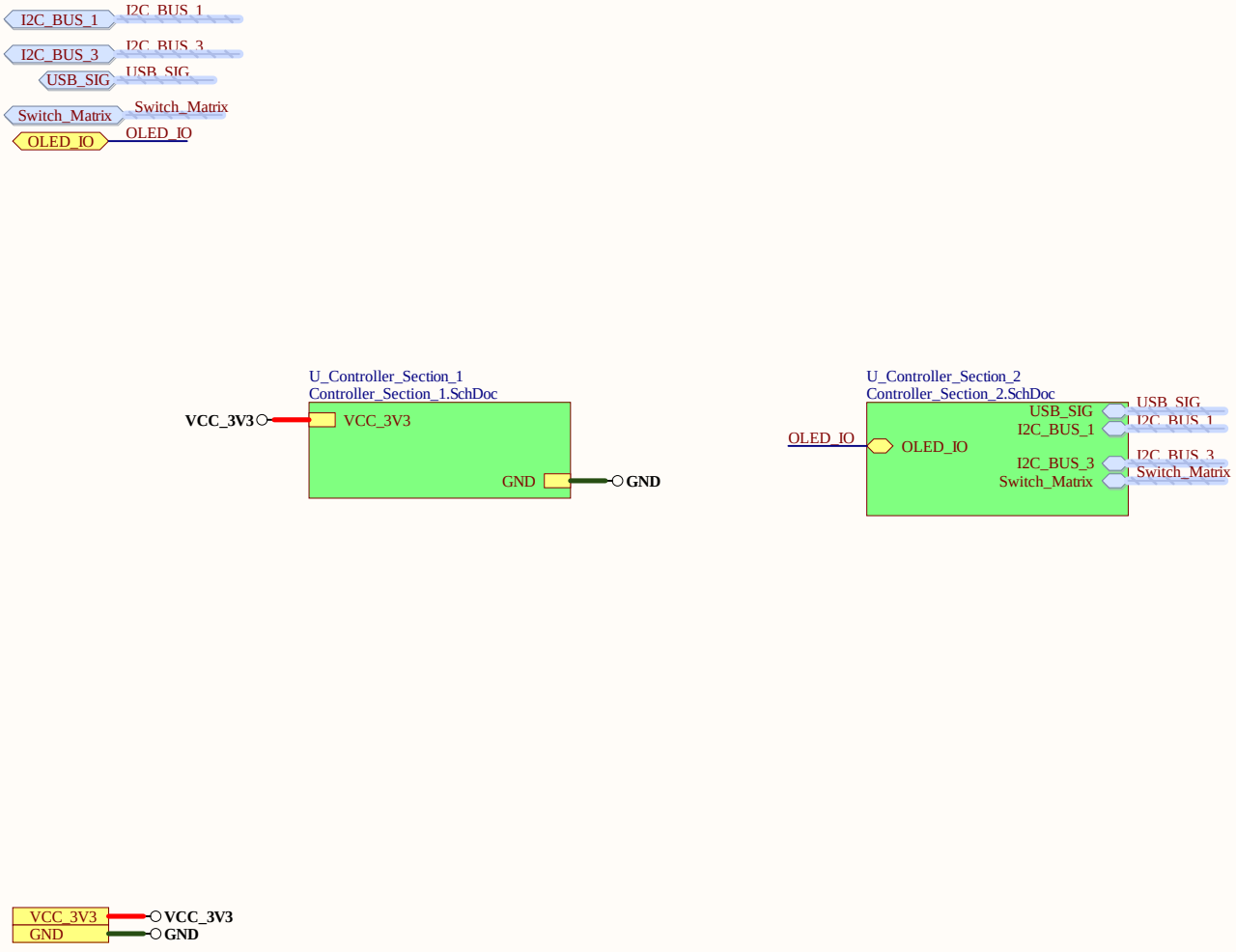

CONTROLLER POWER BANK
VCC_3V3
U2A
11 VDD
19 VDD
28 VDD
50 VDD
75 VDD
100 VDD
VDDA 22 VDDA
6 VBAT
XH_IN 12 PH0-OSC_IN
XH_OUT 13 PH1-OSC_OUT
XL_IN 8 PC14-OSC32_IN
XL_OUT 9 PC15-OSC32_OUT
BOOT0 94 BOOT0
PB2/BOOT1 37 BOOT1
VCAP_1 49 C14 2.2uF
VCAP_2 73 C15 2.2uF
VSSA/VREF- 20
VREF+ 21 VREF
NRST 14 NRST
VSS 10
VSS 27
VSS 74
VSS 99
STM32F407VGT7TR
GND


HSE & LSE CRYSTAL
XL_IN X1 32.768KHz XL_OUT
C16 10pF
C17 10pF
GND
GND
XH_IN X2 25MHz XH_OUT
C19 10pF
C20 10pF
GND
GND
GND
GND


BOOT MODE SELECTOR
VCC_3V3
SW1
1
2 R2 10K BOOT0
3
SS-3235D-03-L2
GND
VCC_3V3
SW2
1
2 R3 10K BOOT1
3
SS-3235D-03-L2
GND

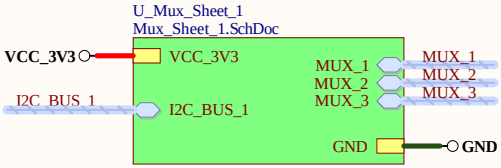

RESET SWITCH
NRST SW3
C18 0.1uF
GND

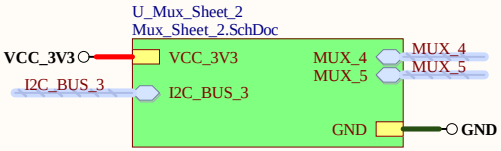

VCC_3V3
GND
VCC_3V3
GND

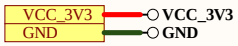

I2C_BUS_1
I2C_BUS_1
I2C_BUS_3
I2C_BUS_3
USB_SIG
USB_SIG
Switch_Matrix
Switch_Matrix
OLED_IO
OLED_IO
U_Controller_Section_1
Controller_Section_1.SchDoc
VCC_3V3
VCC_3V3
GND
GND
U_Controller_Section_2
Controller_Section_2.SchDoc
OLED_IO
OLED_IO
USB_SIG
USB_SIG
I2C_BUS_1
I2C_BUS_1
I2C_BUS_3
I2C_BUS_3
Switch_Matrix
Switch_Matrix
VCC_3V3
VCC_3V3
GND
GND

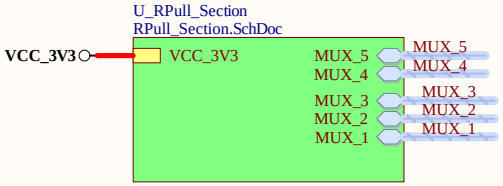

I2C_BUS_1
I2C_BUS_1
I2C_BUS_3
I2C_BUS_3

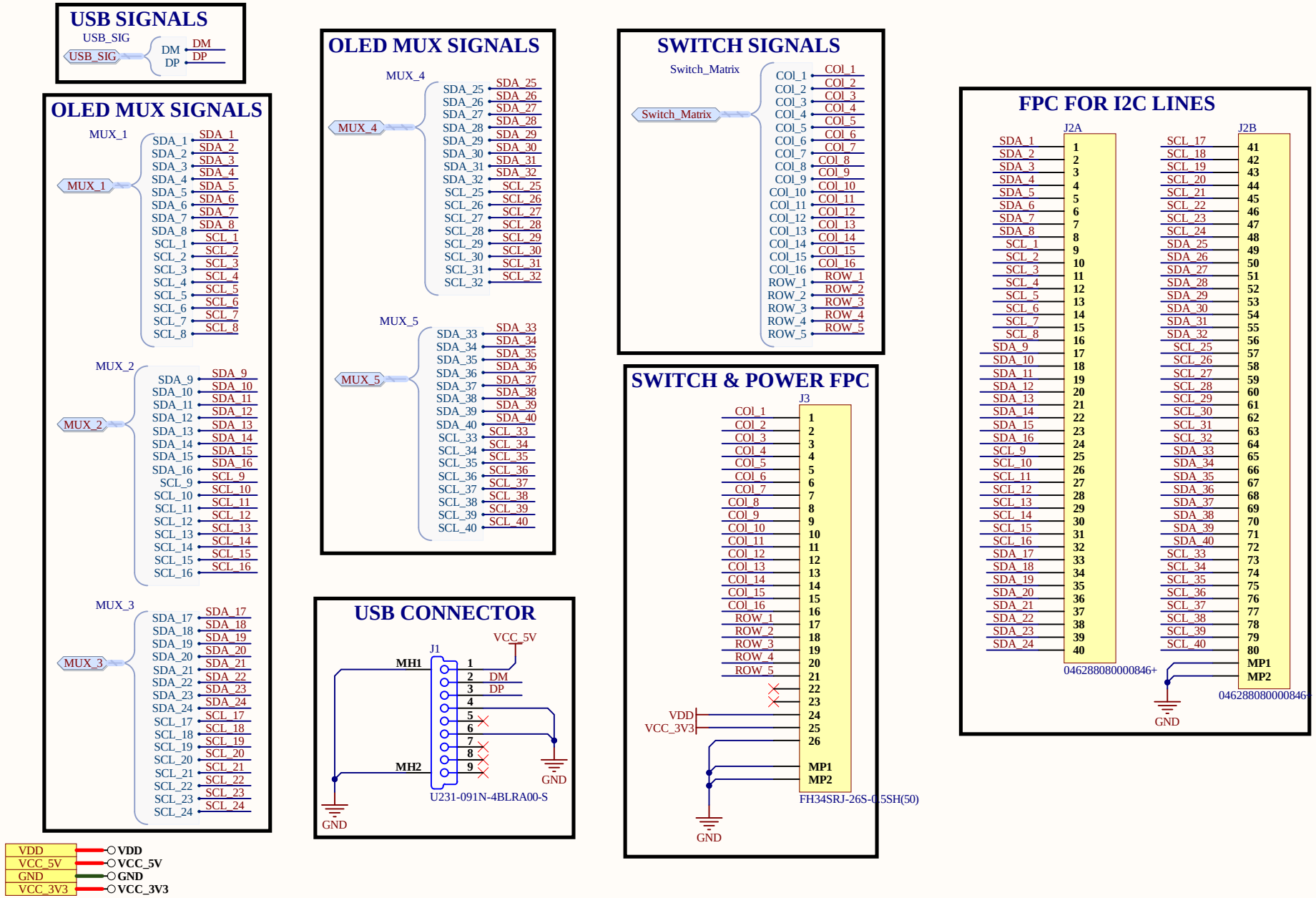

MUX_1
MUX_2
MUX_3
MUX_4
MUX_5
MUX_1
MUX_2
MUX_3
MUX_4
MUX_5

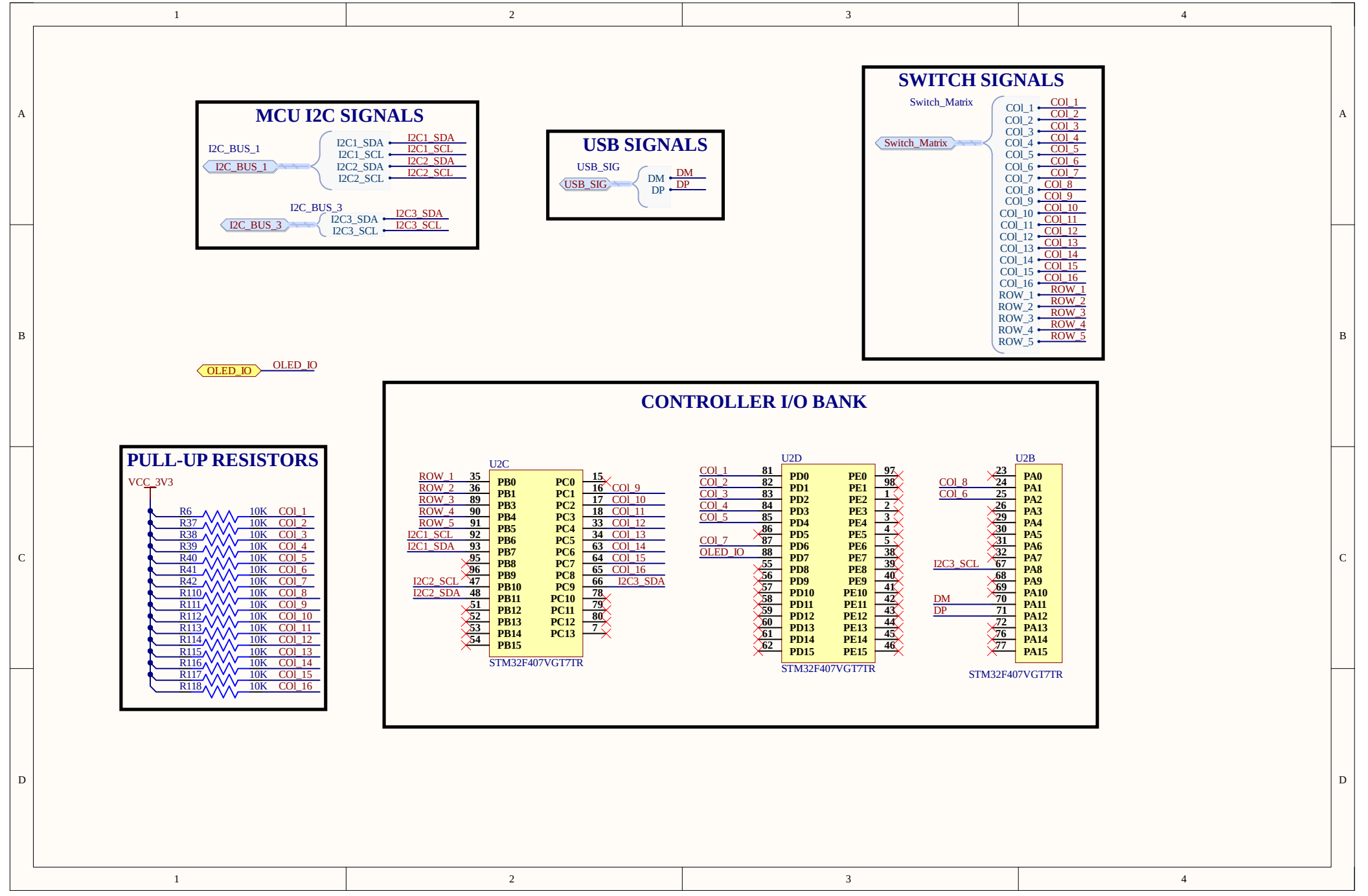

U_Mux_Sheet_1
Mux_Sheet_1.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_1
I2C_BUS_1
MUX_1
MUX_2
MUX_3
MUX_1
MUX_2
MUX_3
GND
GND

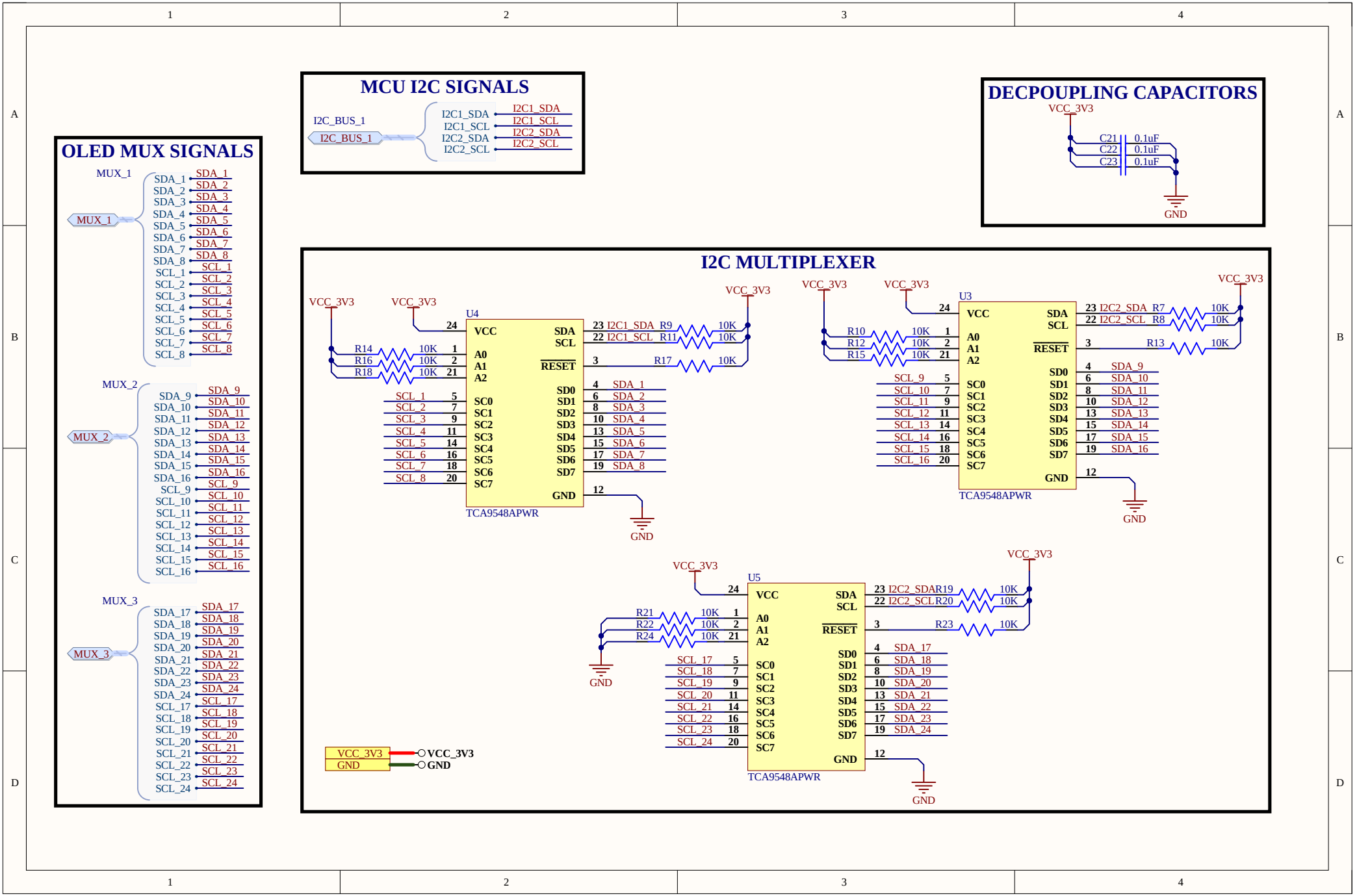

U_Mux_Sheet_2
Mux_Sheet_2.SchDoc
VCC_3V3
VCC_3V3
I2C_BUS_3
I2C_BUS_3
MUX_4
MUX_5
MUX_4
MUX_5
GND
GND

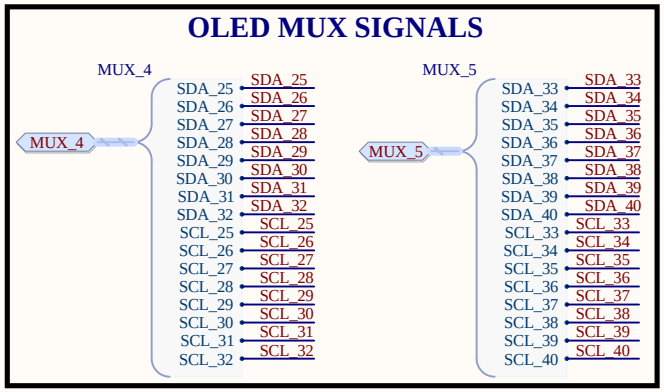

VCC_3V3
GND
VCC_3V3
GND

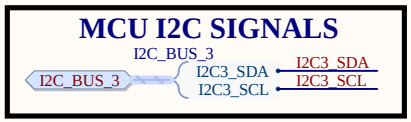

U_RPull_Section
RPull_Section.SchDoc
VCC_3V3
VCC_3V3
MUX_5
MUX_4
MUX_3
MUX_2
MUX_1
MUX_5
MUX_4
MUX_3
MUX_2
MUX_1

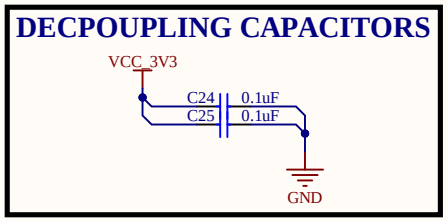

USB SIGNALS
USB_SIG
DM
DP
OLED MUX SIGNALS
MUX_1
MUX_2
MUX_3
OLED MUX SIGNALS
MUX_4
MUX_5
SWITCH SIGNALS
Switch_Matrix
SWITCH & POWER FPC
J3
FH34SRJ-26S-0.5SH(50)
FPC FOR I2C LINES
J2A
J2B
0462880800008464
USB CONNECTOR
J1
MH1
MH2
U231-091N-4BLRA00-S
VCC_5V
GND
VDD
VCC_3V3

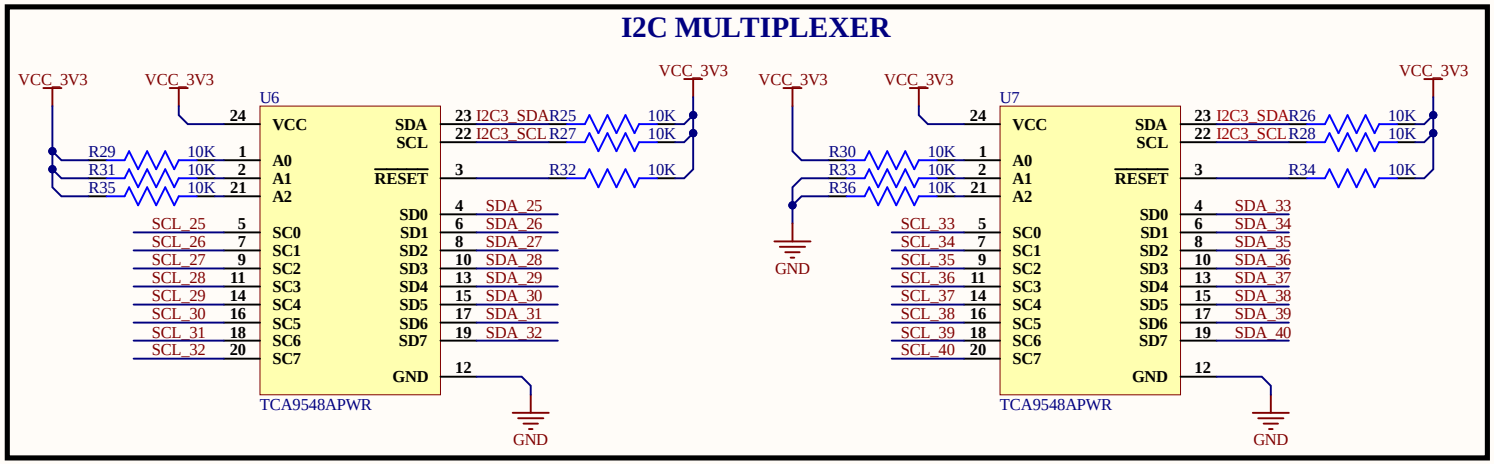

MCU I2C SIGNALS
I2C_BUS_1
I2C_BUS_1
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
I2C_BUS_3
I2C_BUS_3
I2C3_SDA
I2C3_SCL
I2C3_SDA
I2C3_SCL
USB SIGNALS
USB_SIG
USB_SIG
DM
DP
DM
DP
SWITCH SIGNALS
Switch_Matrix
Switch_Matrix
COl_1
COl_2
COl_3
COl_4
COl_5
COl_6
COl_7
COl_8
COl_9
COl_10
COl_11
COl_12
COl_13
COl_14
COl_15
COl_16
ROW_1
ROW_2
ROW_3
ROW_4
ROW_5
OLED_IO
OLED_IO
PULL-UP RESISTORS
VCC_3V3
R6 10K COl_1
R37 10K COl_2
R38 10K COl_3
R39 10K COl_4
R40 10K COl_5
R41 10K COl_6
R42 10K COl_7
R110 10K COl_8
R111 10K COl_9
R112 10K COl_10
R113 10K COl_11
R114 10K COl_12
R115 10K COl_13
R116 10K COl_14
R117 10K COl_15
R118 10K COl_16
CONTROLLER I/O BANK
U2C
ROW_1 35 PB0 PC0 15 COl_9
ROW_2 36 PB1 PC1 16 COl_10
ROW_3 89 PB3 PC2 17 COl_11
ROW_4 90 PB4 PC3 18 COl_12
ROW_5 91 PB5 PC4 33 COl_13
I2C1_SCL 92 PB6 PC5 34 COl_14
I2C1_SDA 93 PB7 PC6 63 COl_15
95 PB8 PC7 64 COl_16
96 PB9 PC8 65 I2C3_SDA
I2C2_SCL 47 PB10 PC9 66
I2C2_SDA 48 PB11 PC10 78
51 PB12 PC11 79
52 PB13 PC12 80
53 PB14 PC13 7
54 PB15
STM32F407VGT7TR
U2D
COl_1 81 PD0 PE0 97
COl_2 82 PD1 PE1 98
COl_3 83 PD2 PE2 1
COl_4 84 PD3 PE3 2
COl_5 85 PD4 PE4 3
86 PD5 PE5 4
COl_7 87 PD6 PE6 5
OLED_IO 88 PD7 PE7 38
55 PD8 PE8 39
56 PD9 PE9 40
57 PD10 PE10 41
58 PD11 PE11 42
59 PD12 PE12 43
60 PD13 PE13 44
61 PD14 PE14 45
62 PD15 PE15 46
STM32F407VGT7TR
U2B
COl_8 23 PA0
COl_6 24 PA1
25 PA2
26 PA3
29 PA4
30 PA5
31 PA6
32 PA7
I2C3_SCL 67 PA8
68 PA9
69 PA10
DM 70 PA11
DP 71 PA12
72 PA13
76 PA14
77 PA15
STM32F407VGT7TR


OLED MUX SIGNALS
MUX_1
MUX_1
SDA_1 SDA_2 SDA_3 SDA_4 SDA_5 SDA_6 SDA_7 SDA_8
SCL_1 SCL_2 SCL_3 SCL_4 SCL_5 SCL_6 SCL_7 SCL_8
MUX_2
MUX_2
SDA_9 SDA_10 SDA_11 SDA_12 SDA_13 SDA_14 SDA_15 SDA_16
SCL_9 SCL_10 SCL_11 SCL_12 SCL_13 SCL_14 SCL_15 SCL_16
MUX_3
MUX_3
SDA_17 SDA_18 SDA_19 SDA_20 SDA_21 SDA_22 SDA_23 SDA_24
SCL_17 SCL_18 SCL_19 SCL_20 SCL_21 SCL_22 SCL_23 SCL_24
MCU I2C SIGNALS
I2C_BUS_1
I2C_BUS_1
I2C1_SDA I2C1_SCL I2C2_SDA I2C2_SCL
DECPOUPLING CAPACITORS
VCC_3V3
C21 0.1uF
C22 0.1uF
C23 0.1uF
GND
I2C MULTIPLEXER
U4
U3
U5
TCA9548APWR
VCC SDA SCL A0 A1 A2 RESET SC0 SC1 SC2 SC3 SC4 SC5 SC6 SC7 SD0 SD1 SD2 SD3 SD4 SD5 SD6 SD7 GND
I2C1_SDA_R9
I2C1_SCL_R11
I2C2_SDA_R7
I2C2_SCL_R8
I2C2_SDAR19
I2C2_SCLR20
R14 R16 R18 R17 R10 R12 R15 R13 R21 R22 R24 R23
10K
VCC_3V3
GND

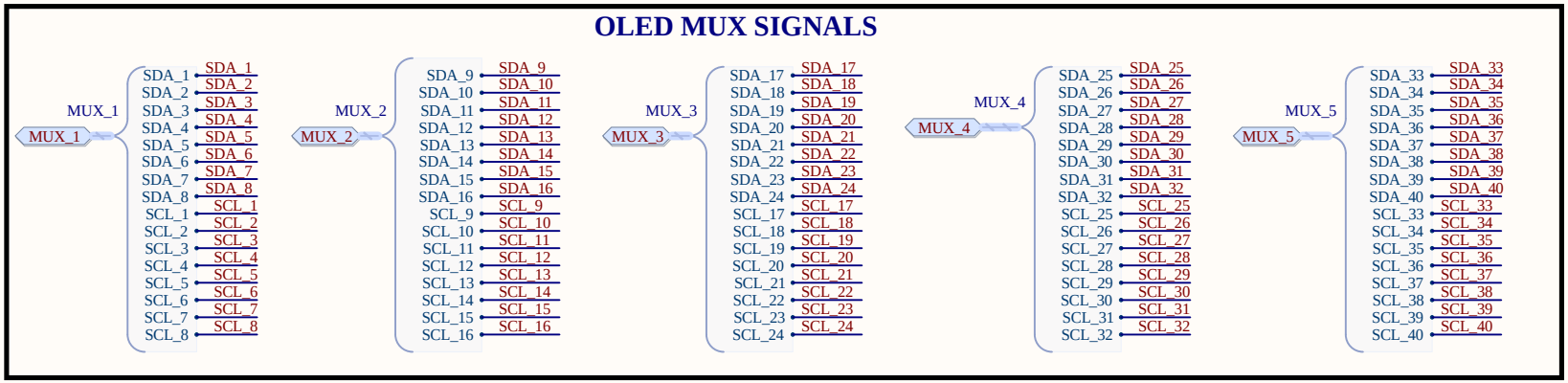

OLED MUX SIGNALS
MUX_4
MUX_4
SDA_25 SDA_25
SDA_26 SDA_26
SDA_27 SDA_27
SDA_28 SDA_28
SDA_29 SDA_29
SDA_30 SDA_30
SDA_31 SDA_31
SDA_32 SDA_32
SCL_25 SCL_25
SCL_26 SCL_26
SCL_27 SCL_27
SCL_28 SCL_28
SCL_29 SCL_29
SCL_30 SCL_30
SCL_31 SCL_31
SCL_32 SCL_32
MUX_5
MUX_5
SDA_33 SDA_33
SDA_34 SDA_34
SDA_35 SDA_35
SDA_36 SDA_36
SDA_37 SDA_37
SDA_38 SDA_38
SDA_39 SDA_39
SDA_40 SDA_40
SCL_33 SCL_33
SCL_34 SCL_34
SCL_35 SCL_35
SCL_36 SCL_36
SCL_37 SCL_37
SCL_38 SCL_38
SCL_39 SCL_39
SCL_40 SCL_40

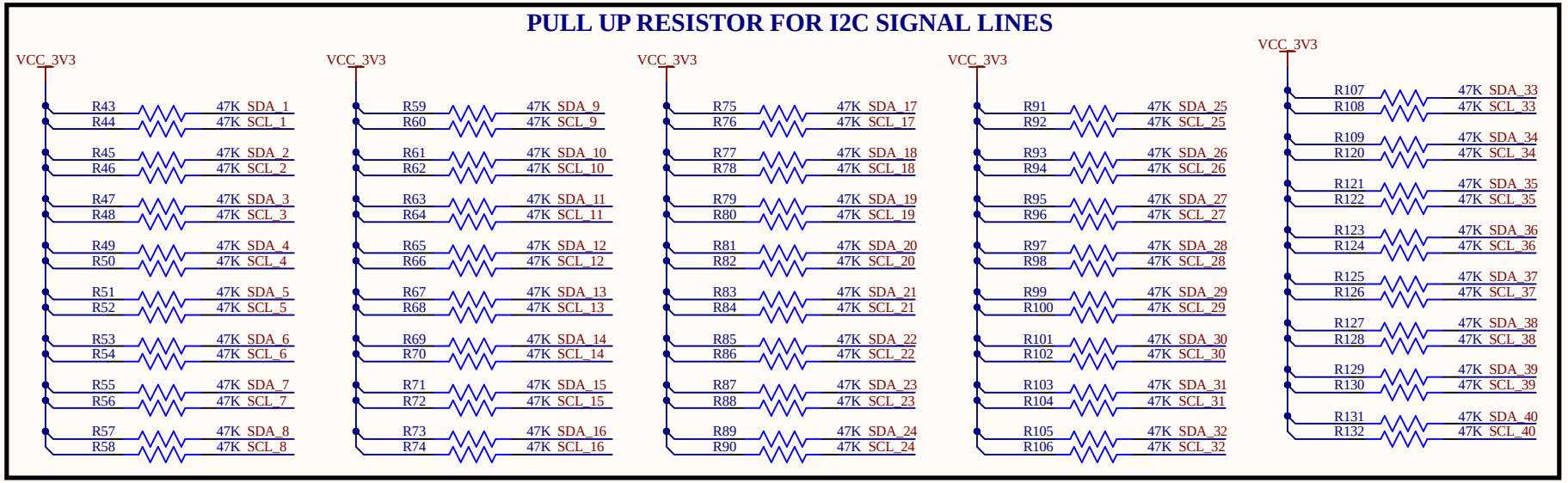

MCU I2C SIGNALS
I2C_BUS_3
I2C_BUS_3
I2C3_SDA
I2C3_SCL
I2C3_SDA
I2C3_SCL


DECPOUPLING CAPACITORS
VCC_3V3
C24 0.1uF
C25 0.1uF
GND

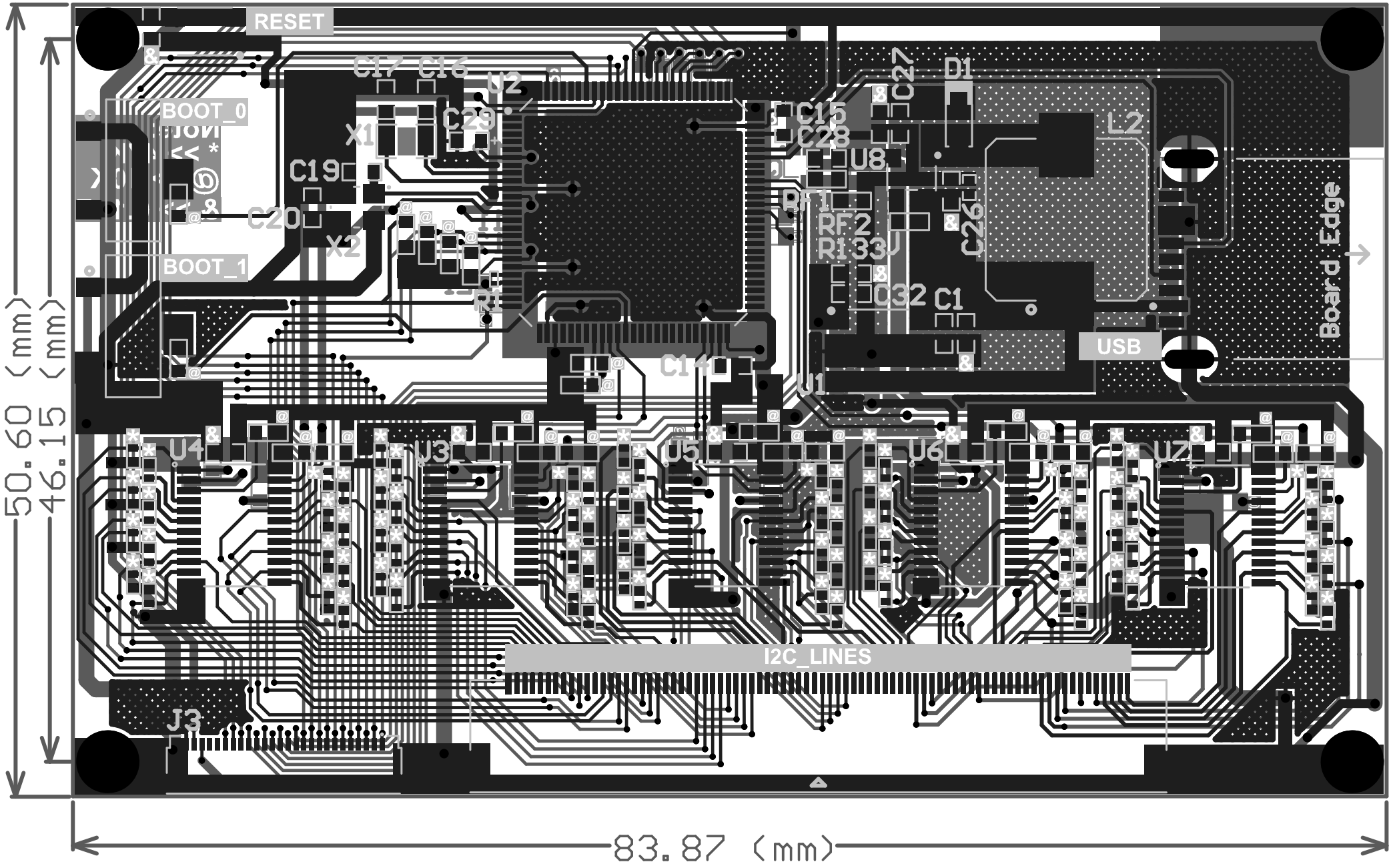

I2C MULTIPLEXER
VCC_3V3
VCC_3V3
U6
VCC
SDA
SCL
RESET
A0
A1
A2
SC0
SC1
SC2
SC3
SC4
SC5
SC6
SC7
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
GND
TCA9548APWR
R29 10K
R31 10K
R35 10K
I2C3_SDAR25
I2C3_SCL R27
R32 10K
10K
10K
SCL_25 SDA_25
SCL_26 SDA_26
SCL_27 SDA_27
SCL_28 SDA_28
SCL_29 SDA_29
SCL_30 SDA_30
SCL_31 SDA_31
SCL_32 SDA_32
GND
VCC_3V3
VCC_3V3
U7
R30 10K
R33 10K
R36 10K
I2C3_SDAR26
I2C3_SCL R28
R34 10K
SCL_33 SDA_33
SCL_34 SDA_34
SCL_35 SDA_35
SCL_36 SDA_36
SCL_37 SDA_37
SCL_38 SDA_38
SCL_39 SDA_39
SCL_40 SDA_40
GND


VCC_3V3
GND
VCC_3V3
GND


OLED MUX SIGNALS
MUX_1
MUX_2
MUX_3
MUX_4
MUX_5
SDA_1 SDA_2 SDA_3 SDA_4 SDA_5 SDA_6 SDA_7 SDA_8 SCL_1 SCL_2 SCL_3 SCL_4 SCL_5 SCL_6 SCL_7 SCL_8
SDA_9 SDA_10 SDA_11 SDA_12 SDA_13 SDA_14 SDA_15 SDA_16 SCL_9 SCL_10 SCL_11 SCL_12 SCL_13 SCL_14 SCL_15 SCL_16
SDA_17 SDA_18 SDA_19 SDA_20 SDA_21 SDA_22 SDA_23 SDA_24 SCL_17 SCL_18 SCL_19 SCL_20 SCL_21 SCL_22 SCL_23 SCL_24
SDA_25 SDA_26 SDA_27 SDA_28 SDA_29 SDA_30 SDA_31 SDA_32 SCL_25 SCL_26 SCL_27 SCL_28 SCL_29 SCL_30 SCL_31 SCL_32
SDA_33 SDA_34 SDA_35 SDA_36 SDA_37 SDA_38 SDA_39 SDA_40 SCL_33 SCL_34 SCL_35 SCL_36 SCL_37 SCL_38 SCL_39 SCL_40


PULL UP RESISTOR FOR I2C SIGNAL LINES
VCC_3V3
R43 R44 R45 R46 R47 R48 R49 R50 R51 R52 R53 R54 R55 R56 R57 R58
47K_SDA_1 47K_SCL_1 47K_SDA_2 47K_SCL_2 47K_SDA_3 47K_SCL_3 47K_SDA_4 47K_SCL_4 47K_SDA_5 47K_SCL_5 47K_SDA_6 47K_SCL_6 47K_SDA_7 47K_SCL_7 47K_SDA_8 47K_SCL_8
R59 R60 R61 R62 R63 R64 R65 R66 R67 R68 R69 R70 R71 R72 R73 R74
47K_SDA_9 47K_SCL_9 47K_SDA_10 47K_SCL_10 47K_SDA_11 47K_SCL_11 47K_SDA_12 47K_SCL_12 47K_SDA_13 47K_SCL_13 47K_SDA_14 47K_SCL_14 47K_SDA_15 47K_SCL_15 47K_SDA_16 47K_SCL_16
R75 R76 R77 R78 R79 R80 R81 R82 R83 R84 R85 R86 R87 R88 R89 R90
47K_SDA_17 47K_SCL_17 47K_SDA_18 47K_SCL_18 47K_SDA_19 47K_SCL_19 47K_SDA_20 47K_SCL_20 47K_SDA_21 47K_SCL_21 47K_SDA_22 47K_SCL_22 47K_SDA_23 47K_SCL_23 47K_SDA_24 47K_SCL_24
R91 R92 R93 R94 R95 R96 R97 R98 R99 R100 R101 R102 R103 R104 R105 R106
47K_SDA_25 47K_SCL_25 47K_SDA_26 47K_SCL_26 47K_SDA_27 47K_SCL_27 47K_SDA_28 47K_SCL_28 47K_SDA_29 47K_SCL_29 47K_SDA_30 47K_SCL_30 47K_SDA_31 47K_SCL_31 47K_SDA_32 47K_SCL_32
R107 R108 R109 R120 R121 R122 R123 R124 R125 R126 R127 R128 R129 R130 R131 R132
47K_SDA_33 47K_SCL_33 47K_SDA_34 47K_SCL_34 47K_SDA_35 47K_SCL_35 47K_SDA_36 47K_SCL_36 47K_SDA_37 47K_SCL_37 47K_SDA_38 47K_SCL_38 47K_SDA_39 47K_SCL_39 47K_SDA_40 47K_SCL_40


VCC_3V3
VCC_3V3


RESET
BOOT_0
BOOT_1
C17
C16
U2
X1
C23
C19
C20
X2
R1
C15
C28
U8
C27
D1
L2
RF1
RF2
R133
C26
C32
C1
C11
USB
Board Edge
U1
U4
U3
U5
U6
U7
I2C_LINES
J3
50.60 (mm)
46.15 (mm)
83.87 (mm)

Comment	Description	Designator	Footprint	LibRef	Quantity
4.7UF	CAP CER 4.7UF 25V X5R 0603	C1, C13, C32	Cap0603	CL10A475KA8NQNC	3
0.1uF	CAP CER 0.1UF 50V Y5V 0603	C2, C3, C4, C5, C6, C7, C8, C9, C10, C18, C21, C22, C23, C24, C25, C30, C31, C33	Cap0603	CC0603ZRY5V9BB104	18
1uF	CAP CER 1UF 25V X5R 0603	C11, C12	Cap0603	CC0603KRX5R8BB105	2
2.2uF	CAP, 2.2µF, 35V, 10%, X5R, 0603	C14, C15	Cap0603	GRM188R6YA225KA12D	2
10pF	CAP CER 10PF 50V NP0 0603	C16, C17, C19, C20	Cap0603	CC0603JRNPO9BN100	4
2.2uF	CAP CER 2.2UF 16V X5R 0402	C26	Cap 0402	EMK105ABJ225KV-F	1
10uF	CAP CER 10UF 25V X5R 0603	C27	Cap0603	C1608X5R1E106M080AC	1
220pF	RES SMD 220pF ±10% 50V X7R 0603	C28	Cap0603	CC0603KRX7R9BB221	1
1uF	CAP TANT 1UF 20% 25V 0603	C29	F981E105MMA	F981E105MMA	1
MBR0520	20V 450mV@500mA 500mA SOD-123 Schottky Barrier Diodes (SBD) ROHS	D1	SOD123	MBR0520	1
U231-091N-4BLRA00-S	USB 3.0 1 9 Male Type-A SMD USB Connectors ROHS	J1	U231-091N-4BLRA00-S	U231-091N-4BLRA00-S	1
046288080000846+	0.5MM PITCH - 80 POS - RIGHT ANG	J2	046288080000846+	046288080000846+	1
FH34SRJ-26S-0.5SH(50)	CONN FFC FPC 26POS 0.50MM R/A	J3	FH34SRJ-26S-0.5SH(50)	FH34SRJ-26S-0.5SH(50)	1
FBMH1608HL121-TV	Ferrite Beads 120Ohm 25% 100MHz 1.2A 0.13Ohm DCR 0603 Automotive T/R	L1	IND 0603	FBMH1608HL121-TV	1
3.3UH	FIXED IND 3.3UH 8A 11.8 MOHM SMD	L2	74437368033	74437368033	1
FDN338P	20V 1.6A 500mW 115mΩ@4.5V,1.6A 1.5V@250µA P Channel SOT-23 MOSFETs ROHS	Q1	SOT23-3	FDN338P	1
FDN335N	SOT-23 MOSFETs ROHS	Q2	SOT23-3	FDN335N	1
0R	RES SMD 0 OHM JUMPER 1/20W 0201	R1	Res-0201	RC0201JR-070RL	1
10K	RES SMD 10K OHM 5% 1/10W 0603	R2, R3, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R110, R111, R112, R113, R114, R115, R116, R117, R118	Res0603	AC0603JR-0710KL	48
47K	RES SMD 47K OHM 1% 1/16W 0402	R4, R5, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100, R101, R102, R103, R104, R105, R106, R107, R108, R109, R120, R121, R122, R123, R124, R125, R126, R127, R128, R129, R130, R131, R132	Res0402	RC0402FR-0747KL	82
51K	Res Thick Film 0603 51K Ohm 1% 0.1W(1/10W) ±100ppm/C Pad SMD Automotive T/R	R133	Res0603	RMCF0603FT51K0	1
68K	RES SMD 68K OHM 1% 1/10W 0603	RF1	Res0603	RC0603FR-0768KL	1
13.3K	RES SMD 13.3K OHM 1% 1/10W 0603	RF2	Res0603	AC0603FR-0713K3L	1
SS-3235D-03-L2	SPDT 50mA @ 12VDC 12V On-On SMD Toggle Switches ROHS	SW1, SW2	SS-3235D-03-L2	SS-3235D-03-L2	2
TS-1185EC-C-D-B	3mm 2mm Round Button 50mA Top3.65mm SPST 12V SMD Tactile Switches	SW3	TS-1185EC-C-D-B	TS-1185EC-C-D-B	1
S-1172B33-E6T1U	IC REG LINEAR 3.3V 1A 6HSOP	U1	HSOP-6	S-1172B33-E6T1U	1
STM32F407VGT7TR	IC MCU 32BIT 1MB FLASH 100LQFP	U2	LQFP-100	STM32F407VGT7TR	1
TCA9548APWR	IC BUS SWITCH 1 X 8:1 24TSSOP	U3, U4, U5, U6, U7	24-TSSOP	TCA9548APWR	5
LM27313XMF/NOPB	IC REG BOOST ADJ 0.8A SOT23-5	U8	SOT23-5	LM27313XMF/NOPB	1
32.768KHz	32.768kHz SMD Crystal Resonator 7pF 70kΩ ±20ppm	X1	Q13FC1350000200	Q13FC1350000200	1
25MHz	25MHz SMD Crystal Resonator 10pF 40Ω ±10ppm	X2	S3225000101040	S3225000101040	1